Hampus Serneke

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EDUCATION

École polytechnique fédérale de Lausanne, EPFL **Lausanne, Switzerland**
M.Sc. in Data Science Sept. 2026 – July 2028

- **Ongoing coursework**: machine learning, mathematics of data, measure theory, foundations of data science, ethics and law of AI, French

Lund University **Lund, Sweden**
B.Sc. in Technology as part of the Engineering Mathematics program Aug. 2022 – June 2026

- **Cumulative GPA:** 4.9/5.0
- Pursued a major in Computer Science and Engineering for two years before transferring to Engineering Mathematics.
- **Thesis title:** Deep Learning for Rotation Averaging in Structure from Motion (GNNs in computer vision)
- **Coursework:** Mathematical statistics, matrix theory, stationary stochastic processes, applied mathematics (PDEs), algorithms and data structures, functional programming, object-oriented modelling, electromagnetic field theory

Lund University **Lund, Sweden**
Theoretical Philosophy, 30 ECTS Aug. 2021 – Jan. 2022

- **GPA:** VG/VG
- **Coursework:** Philosophy of science, philosophy of mind, philosophical method, history of philosophy
- Studied philosophy during a gap year while working part-time as a substitute teacher

Katedralskolan Lund **Lund, Sweden**
Natural Science High School program Aug. 2018 – June 2021

- **GPA:** 22.5/22.5

WORK & LEADERSHIP EXPERIENCE

California Institute of Technology, Caltech **Pasadena, USA**
Summer Undergraduate Research Fellow June 2026 – Aug. 2026

- Conducted research in reinforcement learning and AI safety at the Advanced Mechanical Bipedal Experimental Robotics Lab under the supervision of Dr. Pol Mestres and Prof. Aaron Ames
- Created the first hardware implementation of an anytime safe reinforcement learning algorithm (RL-SGF), going from theorem to hardware
- Ran hardware experiments on G1 humanoids and Crazyflie quadrotors using policies trained in IsaacLab and mjlab
- Performed system identification for Crazyflie quadrotors
- Implemented a novel learning rate scheduler for fulfilling the safety requirements, increasing mean reward by 45%
- Obtained a sharper theoretical bound for the required learning rate to ensure safety
- Investigated the effects of relaxing the theory by introducing PPO-inspired estimates

Lund University **Lund, Sweden**
Teaching Assistant in Mathematics, Faculty of Engineering Aug. 2025 – Mar. 2026

- Led weekly exercise sessions where I taught calculus to a class of 30 engineering undergraduates

Axis Communications **Lund, Sweden**
Machine Learning Intern, Core Technologies Analytics June 2023 – May 2025

- Reduced bias in computer vision models using synthetic data
- Developed a pipeline for generating synthetic data in Unreal Engine and independently designed experiments for testing its effectiveness
- Trained deep learning models for object detection and improved overall mAP, while reducing coverage bias by greatly improving performance on key object classes
- Implemented a novel noise reduction algorithm for pre-processing synthetic data in Python
- Streamlined the rendering pipeline in Unreal and refactored the codebase in C++
- Developed a Python package for functional handling of training result files
- Created a visualization tool in Tkinter
- Collaborated across departments to generate data for different applications, onboarded consultants, and worked directly with a team of 10 engineers
- Worked part-time during the terms and full-time during the summers

Hemkodat **Sweden**
Coding Instructor Sept. 2018 – Jan. 2023

- Travelled in Sweden during holidays and weekends to deliver courses in Python, Lua, and C# for adolescents
- Contributed to onboarding and training of new instructors

Internationella Engelska Skolan **Lund, Sweden**
Substitute Teacher Sept. 2018 – Jan. 2022
- Worked part-time as a substitute teacher and taught in English.

The Computer Engineering Guild at LTH **Lund, Sweden**
Vice President Jan. 2023 – Dec. 2023
- Led the board of a student guild with 1,000 members, chaired board meetings, and managed administration

ARKAD Career fair **Lund, Sweden**
Company Host Oct. 2022 – Nov. 2022
- Was responsible for two companies at the career fair (Gapwaves and Axis Communications)

SKILLS, PROJECTS & OTHER

Technical Skills: Python (PyTorch), Java, C++, C#, Scala, Haskell, SQL, CI/CD, Git, ROS2, Isaac Lab

Languages: Swedish native, English C2 (IELTS Academic 8.5/9.0), Spanish A2, learning French

Personal Projects: Created a VR simulation of a student dorm in Unreal, built an automated candy dispenser with a Raspberry Pi, developed an algorithm for solving Ricochet robots in Java, built an NLP model with Keras, developed regression models for biological experiments

Open-Source: Working on an implementation of a more stable geodesic distance loss function in the PyRoMA framework for geometric deep learning with PyTorch

Media coverage: Dalslänningen, Helsingborgs Dagblad

Awards: Won the national qualifiers and received a bronze medal at the European Union Science Olympiad (2019)